ML, Nov. 8th 2019. Sila.

Extra Note on the “Hinge Loss”-function.

Hinge Loss function

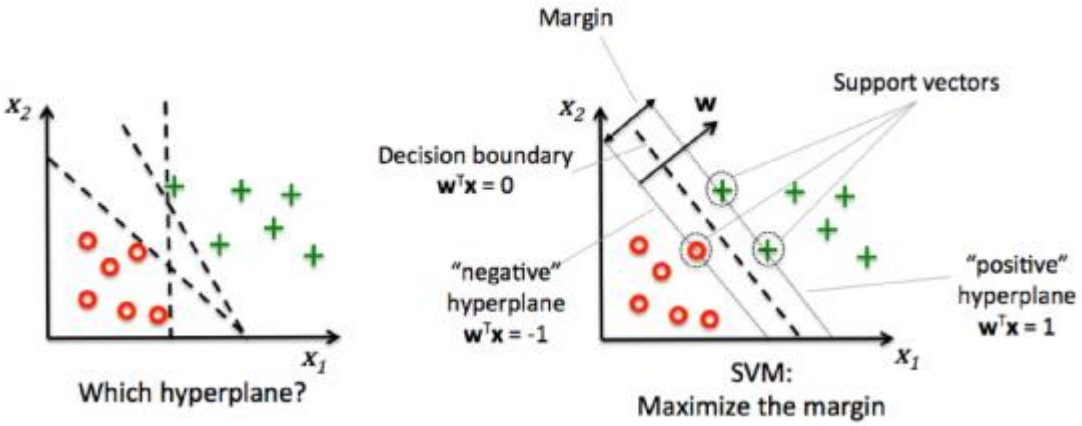
We’ll use the Hinge loss. This is a loss function used for training classifiers. The hinge loss is used for “maximum-margin” classification, most notably for support vector machines (SVMs).

$$c(x, y, f(x)) = (1 - y * f(x))_+$$

c is the loss function, x the sample, y is the true label, f(x) the predicted label.

This means the following: $c(x, y, f(x)) = \begin{cases} 0, & \text{if } y * f(x) \geq 1 \\ 1 - y * f(x), & \text{else} \end{cases}$

SVM. Hinge Loss.



From the diagram,

We want to maximize the distance between positive and negative points. (let's say the distance between the optimal hyperplane to both positive and negative is 1)

So maximize $2/\|w\|$

In other words, we can write this minimize $\|w\|/2$

This is obviously subject to the constraints $y_i(w^T x_i+b)>=1-e_i$

e_i

are the errors which are the distance from the correct hyperplane in the incorrect direction. That is why $e_i s>=0$

(another constraint)

Thus the optimization equation becomes:

$\|w\|/2+1/n\times(e_i)\hat{}2$

We will get the same thing using hinge loss!

Hinge loss:

$max(1-Z_i),0)$

,where$Z_i=y_i(f(x_i))$

Thus,Z_i

is $y_i(w^T x_i+b)$

from earlier condition we know,

$y_i(w^T x_i+b)>=1-e_i$

$Z_i>=1-e_i$

or$e_i>=1-Z_i$

and$e_i>=0$,

combining these two we get

So e_i

is max$(1-Z_i,0)$

Thus $\max(1-Z_i,0)$

is nothing but e_i,These are zero when correctly classified and $(1-Z_i)$

when incorrectly classified!

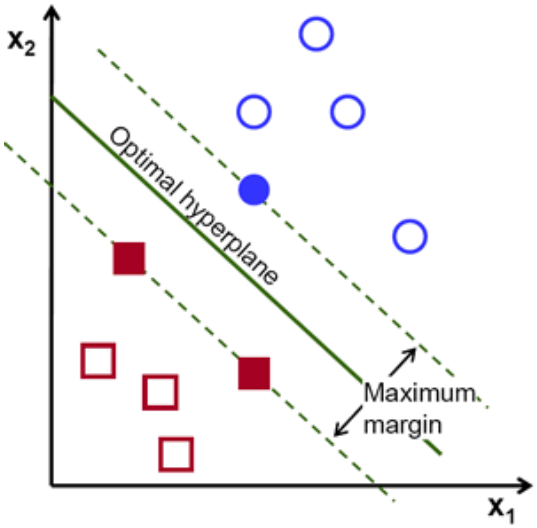
Thus two loss Functions are same which are often written as below:

$min(\|w\|/2+C_{1/n}\times(e_i)\text{^}2)$

$min(max(1-Z_i,0)+\lambda\|w\|)$

And the C can be supposed to be $1/\lambda$

Thus SVMs try to minimise hinge loss!



The hinge loss penalizes predictions not only when they are incorrect, but even when they are correct but not confident. It penalizes gravely wrong predictions significantly, correct but not confident predictions a little less, and only confident, correct predictions are not penalized at all. Let's formalize this by writing out the hinge loss in the case of binary classification:

$\sum_i \max(0,1-y_i*h_\theta(x_i))$

Our labels y_i

are either -1 or 1, so the loss is only zero when the signs match and $|(h_\theta(x_i))|\geq 1$. For example, if our score for a particular training example was 0.2 but the label was −1, we'd incur a penalty of 1.2, if our score was −0.7 (meaning that this instance was predicted to have label −1) we'd still incur a penalty of 0.3, but if we predicted −1.1 then we would incur no penalty.